SAKARYA UNIVERSITY · COP31 GREEN ZONE

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yeşilkod

Measures software energy, fixes the waste, proves the gain.

Yazılımın enerjisini ölçer, israfı düzeltir, kazancı kanıtlar. (“Yeşil Kod” is Turkish for “Green Code”.)

How much electricity does this line of code use? Bu satır ne kadar elektrik harcıyor?

```
- return [no for no in GELENLER if no in kayitlilar]
+ kayitlilar_kume = set(kayitlilar)
+ return [no for no in GELENLER if no in kayitlilar_kume]
```

465× less energy, identical output.

At one million requests a day, this single line emitted about 41 kg CO_2e a year. The engine rewrote it by itself.

An attendance check: every student number was searched in a 6,000-item list. A set makes each lookup constant-time.

HOW IT WORKS

1 Find
Bul

Static analysis walks the code’s syntax tree and flags nine energy-wasting patterns, line by line.

2 Fix
Düzelt

The engine rewrites four of them itself: batched query, set lookup, loop hoisting, caching. Comments and formatting stay untouched.

3 Verify
Doğrula

Both versions are run. If the outputs are not identical, the fix is rejected.

4 Measure
Ölç

CPU energy counters are read; the energy for the same work is converted to carbon.

RESULTS

Four real-world wastes. The engine fixed all four.

None of the optimised code was written by hand.

Route count uncached double recursion → lru_cache	**~4,000×**
Attendance check list lookup inside a loop → set()	**465×**
Price list loop-invariant work → moved out of the loop	**51×**
Order report N+1 queries → one batched IN query	**4,001 → 2 queries**

Reduction in CPU energy per request; estimated measurement mode, 26 October 2026. Ratios vary between machines; the stand panel measures live every time.

Data-centre electricity use more than doubles in six years.

Global data centres, IEA base case.



Source: IEA, Energy and AI (2025). The 2030 value is the base-case projection.

Carbon accounting

Software Carbon Intensity, ISO/IEC 21031:2024

$$SCI = (E \times I + M) / R$$

- **E** measured CPU energy
- **I** Türkiye grid, 341 gCO_2e/kWh
- **M** embodied hardware emissions (next phase)
- **R** per request

Memory, network and cooling are excluded, so the gain shown is a lower bound. Grid data: Low-Carbon Power, Aug 2025–Jul 2026.

SCAN ME

Try your own code on your phone.

Paste Python or pick an example, then measure both versions on your own phone. Your code never leaves your device.

→ yes3760.github.io/yesil-kod

Open source
Açık kaynak kod

Green Coding and Digital Carbon Footprint Optimisation Tool
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Sources: IEA, Energy and AI (2025)
Low-Carbon Power, Türkiye (2025–2026)
Green Software Foundation, SCI